

LA HW

Included exercises: 1, 3, 5, 9, 13, 17, 29, 31, 33, 35, 41, 43, 47

Exercises 1-12

In Exercises 1–12, a matrix A and its characteristic polynomial are given. Find, if possible, an invertible matrix P and a diago-

Original Exercise 1

$$\begin{bmatrix} 7 & 6 \\ -1 & 2 \end{bmatrix}$$

$$(t - 4)(t - 5)$$

Original Exercise 3

$$\begin{bmatrix} 8 & 9 \\ -4 & -4 \end{bmatrix}$$

$$(t - 2)^2$$

$$\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

Original Exercise 5

$$\begin{bmatrix} 3 & 2 & -2 \\ -8 & 0 & -5 \\ -8 & -2 & -3 \end{bmatrix}$$

$$-(t + 5)(t - 2)(t - 3)$$

$$\begin{bmatrix} 2 & 5 & 6 \end{bmatrix}$$

Original Exercise 9

$$\begin{bmatrix} 1 & -2 & 2 \\ 8 & 11 & -8 \\ 4 & 4 & -1 \end{bmatrix}$$

$$-(t - 5)(t - 3)^2$$

$$\begin{bmatrix} -1 & 0 & 0 & 0 \end{bmatrix}$$

Exercises 13-20

In Exercises 13–20, a matrix A is given. Find, if possible, an invertible matrix P and a diagonal matrix D such that $A =$

Original Exercise 13

$$\begin{bmatrix} 16 & -9 \\ 25 & -14 \end{bmatrix}$$

Original Exercise 17

$$\begin{bmatrix} -1 & 2 & -1 \\ 0 & -3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

Exercises 29-48

In Exercises 29–48, determine whether the statements are true or false.

Original Exercise 29

ments are true or false.

Every $n \times n$ matrix is diagonalizable.

Original Exercise 31

If P is an invertible $n \times n$ matrix and D is a diagonal $n \times n$ matrix such that $A = PDP^{-1}$, then the columns of

Original Exercise 33

diagonal entries of D .

If A is a diagonalizable matrix, then there exists a unique

Original Exercise 35

Every diagonalizable $n \times n$ matrix has n distinct eigen-

Original Exercise 41

A diagonal $n \times n$ matrix has n distinct eigenvalues.

Original Exercise 43

The standard vectors are eigenvectors of a diagonal

Original Exercise 47

If the characteristic polynomial of a matrix A factors into a product of linear factors, then A is diagonalizable.